

On the Limits of Quantum Multiparty Simultaneous Communication

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Abstract

The Simultaneous Message Passing (SMP) model provides a fundamental framework for comparing the relative power of classical and quantum communication. In the two-player setting, Gavinsky et al. (STOC 2006) established the incomparability of shared randomness and quantum communication using INDEX COORDINATION, where public-coin protocols need only $O(\log n)$ bits while bounded-error quantum protocols require $\Omega(n^{1/3})$ qubits.

In this work, we establish a multiparty exponential separation by introducing $IC_{k,n}$, a natural k -party generalization of INDEX COORDINATION. We show that public-coin protocols solve $IC_{k,n}$ unambiguously with maximum message length $O(\log n)$. In contrast, the maximum message length of every quantum SMP protocol without pre-shared entanglement or public coins is $\Omega(n^{1-1/k})$ qubits in the unambiguous regime and $\Omega(n^{\frac{k-1}{k+1}})$ qubits in the general bounded-error regime. The unambiguous lower bound is matched by a classical private-coin protocol with maximum message length $O(n^{1-1/k})$, uniformly in k . For fixed error parameters, the constants in these bounds are independent of k . Consequently, the exponential separation in maximum message length holds for every integer-valued function $k = k(n) \geq 2$, without any restriction on its growth rate. Moreover, both quantum lower bounds become $\Omega(n)$ when $k \geq c \log n$ for any fixed $c > 0$, matching the full-input protocol and establishing tight linear complexity in both error regimes.

Conceptually, our results demonstrate that the coordination afforded by public randomness cannot be efficiently simulated by quantum superposition, extending the exponential incomparability of these resources to arbitrary k . To overcome the complexities of bounding success probabilities for multiparty product states in these regimes, we prove an exact factorization theorem for unambiguous quantum state identification, which may be of independent mathematical interest.

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1 Introduction

The Simultaneous Message Passing (SMP) model, introduced by Yao [22], provides a foundational framework for analyzing the communication bottlenecks and resource trade-offs inherent in distributed computing. In this setting, k spatially separated players receive private inputs and must each send a single, independent message to a central referee. The referee, possessing no input of their own, must then evaluate a joint function or relational problem over the players' distributed data.

A central objective in quantum communication complexity is to establish the comparative power of three distinct resources within this model: classical private randomness, classical public randomness (public coins), and unentangled quantum superposition.

In the two-party setting, this hierarchy is well understood. Classically, public randomness is strictly more powerful than private randomness, best exemplified by the EQUALITY function, which requires only $O(1)$ bits with public coins but $\Omega(n^{1/2})$ bits with private coins [1, 13, 16]. In the quantum domain, Buhrman et al. [3] demonstrated that quantum messages can exponentially outperform classical private randomness via quantum fingerprinting [23]. This raises a fundamental, converse question: *Can quantum communication without pre-shared entanglement efficiently simulate the coordination capabilities of classical shared randomness?*

For two players, Gavinsky, Kempe, and de Wolf [6] answered this in the negative for the unambiguous regime. They introduced the relational problem INDEX COORDINATION, demonstrating that while it admits a straightforward $O(\log n)$ public-coin protocol, any unentangled quantum protocol requires $\Omega(n^{1/2})$ communication in the unambiguous setting, where the referee has two possible outcomes: with high probability returns a correct output, or claims ignorance with an special symbol $\perp$. This was later extended to the general bounded-error regime (where we allow an incorrect symbol in a conclusive output with a fixed probability) by Gavinsky, Kempe, Regev, and de Wolf [7], proving an $\Omega(n^{1/3})$ lower bound. Combined with problems where quantum communication outperforms public coins (e.g., HIDDEN MATCHING [2]), these results establish that quantum communication and shared randomness are strictly incomparable in the two-party SMP model.

Moving beyond two players, however, requires careful theoretical consideration. Early foundational work by Gavinsky and Pudlák [8] established exponential separations in non-interactive multiparty communication complexity in a different input model and resource direction. More directly, Gavinsky, Ito, and Wang [5] showed that shared randomness can exponentially outperform unentangled quantum communication in the number-in-hand SMP model: their promise function GP_k has a public-coin protocol using one bit from each player, whereas every quantum protocol has total communication $\Omega(kn^{1-2/k})$. In particular, at least one player must send $\Omega(n^{1-2/k})$ qubits. Thus, the existence of a genuinely multiparty separation is already known.

What remains open in this line of work is whether the original INDEX COORDINATION phenomenon itself admits a natural multiparty extension. While one can trivially add $k - 2$ “dummy” players to the two-party relation, such a construction does not create a coordination bottleneck that depends on all k inputs. We therefore seek a relation that recovers INDEX COORDINATION at $k = 2$, involves every player non-trivially, and becomes progressively harder for unentangled quantum protocols as the number of players grows. This leads to two more specific questions:

► **Question 1.** *Does INDEX COORDINATION admit a natural k -party generalization that involves all players non-trivially and preserves the separation between classical shared randomness and unentangled quantum communication?*

► **Question 2.** *For such a generalization, can one prove quantum communication lower bounds whose exponents improve with k , in both the unambiguous and bounded-error regimes?*

1.1 Our Contributions

In this work, we answer both questions in the affirmative. We introduce a genuine multiparty extension of INDEX COORDINATION and prove separations in both error regimes. The lower bounds on the longest quantum message have exponents that increase with k and tend to one. In particular, we obtain *tight linear quantum communication complexity in both error regimes* when $k \geq c \log n$ for any fixed $c > 0$: both lower bounds become $\Omega(n)$, matching the $O(n)$ protocol that sends the complete inputs and selector. Indeed, in this range the factors $n^{1/k}$ and $n^{2/(k+1)}$ in the denominators of the two bounds are bounded by constants depending only on c . Since the constants in our upper and lower bounds are independent of k , the exponential separation in maximum message length holds for every integer-valued function $k = k(n) \geq 2$, without restricting its growth rate.

To establish these results, we introduce $\text{IC}_{k,n}$, a natural k -party generalization of INDEX COORDINATION. In this relation, the first $k - 1$ players each receive a string $x_j \in \{0, 1\}^n$. The k -th player receives a string $x_k \in \{0, 1\}^n$ alongside a selector string $s \in \{0, 1\}^n$ with exactly $n/2$ positions set to 1. The referee must output a valid tuple $(i, x_1^{(i)}, \dots, x_k^{(i)})$ such that the selector bit $s^{(i)} = 1$.

This problem elegantly captures the essence of multiparty coordination, directly recovering the classic two-player relation exactly for $k = 2$. Without shared randomness, the first $k - 1$ players have no way to agree on a specific valid index i to transmit. All messages must allow the referee to learn the inputs corresponding to a coordinate selected exclusively by P_k , forcing the first $k - 1$ players to send massive amounts of information to guarantee a valid intersection. At a high level, both lower bounds turn this lack of coordination into an information bottleneck. A short quantum message has only limited predictive power, and quantum information bounds forces that power to be spread across all input coordinates. Once a protocol is fixed, we fix a hard distribution among coordinates where the first players collectively reveal little information.

We establish lower bounds on the longest message in any quantum protocol for $\text{IC}_{k,n}$, with exponents that depend explicitly on the number of players. These bounds show that the coordination provided by shared randomness cannot be efficiently simulated by unentangled quantum messages. We prove the separation in two error regimes:

Communication Model	Error Regime	Cost
Classical Public-Coin	Unambiguous & Bounded-Error	$O(\log n)$ bits
Classical Private-Coin	Unambiguous & Bounded-Error	$O(n^{1-1/k})$ bits
Quantum (No pre-shared entanglement)	Bounded-Error	$\Omega\left(n^{\frac{k-1}{k+1}}\right)$ qubits
Quantum (No pre-shared entanglement)	Unambiguous	$\Theta(n^{1-1/k})$ qubits

■ **Table 1** Bounds on the maximum message length for the multiparty $\text{IC}_{k,n}$ relation in the SMP model, for fixed $0 < \varepsilon < 1$ (with $\varepsilon < 1/8$ for the bounded-error quantum lower bound). All implicit constants are independent of k . When $k \geq c \log n$ for any fixed $c > 0$, quantum and classical private-coin complexity is $\Theta(n)$ in both error regimes, while the public-coin upper bound remains $O(\log n)$.

As summarized in Table 1, our public-coin protocol is strictly unambiguous (it never errs, and only aborts with probability at most ε). Against this highly efficient $O(\log n)$ protocol, we present two major contributions:

1. **Bounded-Error Lower Bounds via Parity Collapse:** Extending the separation to k players requires a product bound on the success probability of the referee respect the k local registers of the players. This bound is notoriously hard to establish for k mixed states in the bounded-error setting [12, 14, 19]. We circumvent this by collapsing the information of many players in one bit. By grouping the first $k - 1$ players into a single register defined by the parity of their inputs, we reduce the k -party coordination constraint to a two-register asymmetric state identification task. This allows us to bridge the existing 2-register identification theorems to our setting, proving that at least one player must send $\Omega(n^{(k-1)/(k+1)})$ qubits in any bounded-error protocol without shared randomness or prior entanglement.
2. **Tight Unambiguous Lower Bounds:** To achieve tighter bounds in the unambiguous special case, we generalize to multiparty setting a technical result for unambiguous quantum discrimination. In the line of recent developments on bounds for the optimal local discrimination of multipartite quantum states [10], we prove a bound for the unambiguous identification of mixed product states over arbitrary finite alphabets (Section 3.5). Using our generalized direct-product theorem, we prove that at least one player must send $\Omega(n^{1-1/k})$ qubits in any unambiguous protocol for $\text{IC}_{k,n}$. By employing an information-pooling argument and the Arithmetic Mean-Geometric Mean (AM-GM) inequality, we establish an exponential separation that gets stronger as the number of players k increases.

2 Related Work

Our work resides at the intersection of quantum communication complexity and quantum state discrimination, and builds upon several major historical milestones in the SMP model.

Two-Party Separations

The foundational bounds establishing that quantum communication cannot simulate public coins were proved by Gavinsky, Kempe, Regev, and de Wolf [6, 7] using the two-player INDEX COORDINATION problem. Our work serves as the natural multiparty generalization of this phenomenon. Specifically, our bounded-error proof relies on their foundational 2-register asymmetric identification bounds and random access code lemmas [7] as the mathematical endpoint of our parity-collapse reduction.

Multiparty Shared Randomness vs. Quantum Communication

In the multiparty domain, Gavinsky, Ito, and Wang [5] previously established that shared randomness can exponentially outperform unentangled quantum messages. However, our work differs conceptually and methodologically from theirs. They studied a specific promise function, GP_k (Gap-Parity), and used a hybrid argument over trace distances to prove a total-communication lower bound of $\Omega(kn^{1-2/k})$, which implies that some player must send $\Omega(n^{1-2/k})$ qubits. In contrast, we study a natural relational coordination problem ($\text{IC}_{k,n}$). Because we analyze state identification and product structures directly rather than relying on hybrid arguments, we obtain lower bounds on the maximum message length with exponents $1 - 1/k$ (unambiguous) and $(k - 1)/(k + 1)$ (bounded-error), both of which scale cleanly with k .

Multipartite Entanglement vs. Unentangled Messages

Very recently, Chakraborty, Banik, and de Wolf [4] demonstrated that SMP protocols utilizing shared *multipartite entanglement* (such as GHZ states) can exponentially outperform unentangled quantum communication. It is crucial to distinguish their resource model from ours. Their efficient protocols require active, prior quantum entanglement distributed among the players. Our work, by contrast, strictly studies the limits of unentangled quantum messages attempting to simulate classical shared randomness, confirming that without prior entanglement, quantum communication remains severely bottlenecked by multiparty coordination tasks.

3 Preliminaries

In this section, we establish the foundational definitions, communication models, and information-theoretic tools required for our multiparty separations. For a comprehensive introduction to general quantum information theory, we refer the reader to standard texts such as Nielsen and Chuang [17] and Watrous [21].

3.1 Quantum Information Basics

We restrict our attention to finite-dimensional Hilbert spaces, denoted by $\mathcal{H}$. The set of linear operators on $\mathcal{H}$ is denoted by $L(\mathcal{H})$, and the set of density operators (positive semidefinite operators with unit trace) is denoted by $D(\mathcal{H})$. A quantum state is represented by a density operator $\rho \in D(\mathcal{H})$. If ρ is a rank-one projector, it represents a pure state.

For any operator $A \in L(\mathcal{H})$, the support $\text{supp}(A)$ is the orthogonal complement of its kernel, $\ker(A)$. If $S \subseteq \mathcal{H}$ is a closed subspace, $S^\perp$ denotes its orthogonal complement, and Π_S is the orthogonal projector onto S .

To quantify the distinguishability of two quantum states, we rely on the trace norm. For an operator A , the trace norm is defined as $\|A\|_{\text{tr}} = \frac{1}{2} \text{Tr}(\sqrt{A^\dagger A})$. Under this convention, the trace distance between two quantum states ρ_0 and ρ_1 lies in the interval $[0, 1]$ and perfectly characterizes the maximum bias with which any physical measurement can distinguish the two states.

A quantum measurement is described by a Positive Operator-Valued Measure (POVM), which is a set of positive semidefinite operators $\{E_i\}$ such that $\sum_i E_i = I$. Operationally, if a system is in state ρ , the exact probability of obtaining outcome i upon applying this measurement is precisely $\text{Tr}(E_i \rho)$.

3.2 The (p, η) -Predictor and State Identification

To analyze protocols in the bounded-error regime, we utilize the framework of bounded-error quantum state identification introduced by Gavinsky, Kempe, Regev, and de Wolf [7].

Consider a quantum state α_X that encodes a uniformly distributed random bit $X \in \{0, 1\}$. A referee wishes to guess X but is allowed to occasionally output a conclusive failure symbol, $\perp$, representing ignorance. A POVM with three outcomes $\{0, 1, \perp\}$ is defined as a (p, η) -predictor for X if it satisfies two conditions:

1. It produces a definitive guess with probability exactly $p = \Pr[\text{outcome} \neq \perp]$.
2. Conditioned on making a definitive guess, the probability that the guess is incorrect is at most η . That is, $\Pr[X \neq \text{outcome} \mid \text{outcome} \neq \perp] \leq \eta$.

The quantity $D_\eta(\alpha_0, \alpha_1)$ is defined as the supremum over all possible POVMs of the success probability p such that the measurement is a (p, η) -predictor for X . This metric cleanly

captures the trade-off between the frequency of a measurement producing a conclusive answer and the reliability of that answer.

3.3 Communication Models and Error Regimes

In the multiparty Simultaneous Message Passing (SMP) model, k spatially separated players $P_1, \dots, P_k$ receive private inputs $x_j \in X_j$. They cannot communicate with each other; instead, each player applies a local encoding function and sends a single message to a central referee. The referee, who receives no input, must process these k messages to evaluate a relational problem $R \subseteq X_1 \times \dots \times X_k \times Z$, outputting a solution $z \in Z$. If m_j denotes the worst-case length of player P_j 's message, we measure communication throughout the paper by the maximum message length

$$m := \max_{j \in [k]} m_j,$$

rather than by the sum of all message lengths. Thus, a lower bound on m means that at least one player must send a message of that length.

We evaluate communication costs across three principal resources:

- **Classical Public-Coin** ($R_{\text{pub}}^{\parallel}$): All players (excluding the referee) share access to a common, unbounded random string that does not count toward the communication cost. The messages sent to the referee are classical bit strings.
- **Classical Private-Coin** ($R_{\text{priv}}^{\parallel}$): Players do not share any public randomness, but each player has access to an independent, local source of random coin flips. The messages sent to the referee are classical bit strings. This model is strictly weaker than the unentangled quantum model ($Q^{\parallel}$), because an m -qubit quantum message can perfectly simulate m classical private coin flips.
- **Quantum without Pre-pre-shared entanglement** ($Q^{\parallel}$): Each player P_j sends a quantum state $\rho_j(x_j)$ of dimension 2^{m_j} (i.e., an m_j -qubit message). The referee applies a POVM to the joint state $\bigotimes_{j=1}^k \rho_j(x_j)$. Crucially, the players do not share any prior entanglement or public coins; the global state received by the referee is strictly a product state over the players' subspaces.

We evaluate the success of these protocols under two strict error definitions, governed by a constant parameter $\varepsilon \in [0, 1]$:

- **Bounded-Error**: The referee is allowed to output a special failure symbol $\perp$. The worst-case probability over all inputs that the protocol fails—either by outputting an invalid tuple $(x_1, \dots, x_k, z) \notin R$ or by outputting $\perp$ —is at most ε . Equivalently, the referee outputs a valid tuple with probability at least $1 - \varepsilon$.
- **Unambiguous**: The referee is allowed to output a special failure symbol $\perp$ with probability at most ε . However, the protocol is strictly forbidden from making an error: whenever the referee outputs a definitive answer $z \neq \perp$, the tuple must be valid, meaning $(x_1, \dots, x_k, z) \in R$ with probability 1.

Because unambiguous protocols never output an incorrect answer and only fail by aborting, any valid unambiguous protocol is trivially also a valid bounded-error protocol. Note that while upper bounds must satisfy these criteria in the worst-case, lower bounds are proven by analyzing the average-case error over a specifically chosen hard input distribution.

3.4 Information-Theoretic Tools

To establish lower bounds on quantum communication, we utilize the Random Access Code (RAC) lemma, originally pioneered by Nayak [15] and adapted for bounded-error state

identification by Gavinsky et al. [7]. The RAC lemma bounds the sum of the capacities of individual bit predictors extractable from a single quantum state, and applies uniformly to both the unambiguous and bounded-error regimes.

► **Proposition 1** (Random Access Codes [7, 15]). *Let $X = X_1 \cdots X_n$ be uniformly distributed on $\{0,1\}^n$, and let Q be an m -qubit encoding of X . Suppose that, for every $i \in [n]$, there exists a measurement on Q whose outcome is a (λ_i, η_i) -predictor for X_i , where $0 \leq \eta_i \leq 1/2$. The measurement may depend on i . Then*

$$\sum_{i=1}^n \lambda_i (1 - H(\eta_i)) \leq m,$$

Consequently, this lemma yields three critical geometric constraints:

1. **Zero-Error L_1 Bound:** If every predictor is unambiguous, then $\eta_i = 0$ for all i . Since $H(0) = 0$, the RAC inequality takes the simple form $\sum_{i=1}^n \lambda_i \leq m$.
2. **Fixed-Error L_1 Bound:** If the predictors operate with a fixed error $\eta < 1/2$, then the sum of the success probabilities is bounded linearly by the encoding size: $\sum_{i=1}^n \lambda_i = O_\eta(m)$.
3. **L_2 Trace-Distance Bound:** If the predictors are derived strictly from distinguishing two mixed states ρ_i^0 and ρ_i^1 , operating as $(1, 1/2 - d_i/2)$ -predictors where $d_i = \|\rho_i^0 - \rho_i^1\|_{\text{tr}}$, then the sum of the squared trace distances is bounded by the encoding size: $\sum_{i=1}^n d_i^2 = O(m)$.

Finally, to pool classical capacities and trace distances across multiple players, we repeatedly use the standard Arithmetic Mean-Geometric Mean inequality [20] in the following form, valid for non-negative real numbers $x_1, \dots, x_N$:

$$\prod_{j=1}^N x_j \leq \left(\frac{1}{N} \sum_{j=1}^N x_j \right)^N.$$

3.5 The Direct-Product Theorem

In the multiparty SMP model, the referee frequently receives a composite message in the form of a product state $\rho = \bigotimes_{j=1}^k \rho_j$. To solve the specific distributed relational problem presented in this paper, the referee must identify which global label $b \in \Sigma_1 \times \cdots \times \Sigma_k$ (corresponding to the players' distributed inputs) is encoded within this product state. Our goal in this section is to rigorously upper bound the best possible success probability of such identification task in the unambiguous setting.

Direct-product theorems of this form are well known in two-party communication complexity, typically demonstrating that solving multiple independent instances of a task requires resources that scale linearly with the number of instances. Here, the task is the unambiguous identification of a composite label from a tensor product of mixed states, and the core challenge is to mathematically characterize the optimal measurement strategy under the strict zero-error constraint.

For the two-player setting ($k = 2$), Gavinsky et al. [6] obtained a lower bound by exploiting a simple factorization of the supports of the relevant quantum states. For an arbitrary number of players k and arbitrary finite alphabets Σ_j , such a factorization is far from obvious, as isolating the correct global state requires projecting to a complex global measurement subspace. Building upon the geometric intuition of the binary case, we establish a stronger generalized result. We demonstrate that, despite this complexity, the global subspace relevant for unambiguous identification still factorizes cleanly into a tensor product of local exclusion subspaces. This structural property allows us to formally state

our direct-product theorem, bounding the global success probability strictly by the product of the local success probabilities.

► **Theorem 1** (Unambiguous Product Identification over Finite Alphabets). *For each $j \in [k]$, let $\{\alpha_j^{(a)}\}_{a \in \Sigma_j}$ be an ensemble of mixed states on $\mathcal{H}_j$. Define the optimal single-register unambiguous success probability for identifying state a as:*

$$p_j^{(a)} := \sup_{M_j} \text{Tr} \left(E_a \alpha_j^{(a)} \right),$$

where the supremum is over all unambiguous POVMs $M_j = \{E_x\}_{x \in \Sigma_j} \cup \{E_\perp\}$.

Consider the ensemble of product states $\rho_b = \bigotimes_{j=1}^k \alpha_j^{(b_j)}$ for $b \in \Sigma_1 \times \cdots \times \Sigma_k$. The maximum probability $P(b)$ of a global unambiguous measurement correctly identifying a specific sequence b is bounded by the product of the local probabilities:

$$P(b) \leq \prod_{j=1}^k p_j^{(b_j)}.$$

The proof of this theorem is deferred to Appendix A

To apply this theorem to the SMP model where inputs are drawn from a uniform prior, we bridge the conditional identification probability of a specific sequence to the average capacities constrained by the RAC inequality.

► **Corollary 1** (Average Capacity Bound). *For each register j , let s_j be the optimal average success probability, under the uniform prior on Σ_j , of a single unambiguous POVM that may identify any symbol:*

$$s_j := \sup_{M_j} \frac{1}{|\Sigma_j|} \sum_{a \in \Sigma_j} \text{Tr} \left(E_a \alpha_j^{(a)} \right),$$

where $M_j = \{E_a\}_{a \in \Sigma_j} \cup \{E_\perp\}$ is required to be unambiguous for the entire ensemble. For any specific target sequence $b \in \Sigma_1 \times \cdots \times \Sigma_k$, the global unambiguous success probability $P(b)$ is bounded by:

$$P(b) \leq \left(\prod_{j=1}^k |\Sigma_j| \right) \prod_{j=1}^k s_j.$$

For binary alphabets where $|\Sigma_j| = 2$ for all j , this reduces to $P(b) \leq 2^k \prod_{j=1}^k s_j$.

Proof. Fix a register j and a target symbol a . A one-sided POVM attaining, or arbitrarily approaching, $p_j^{(a)}$ in Theorem 1 is also a valid POVM for the average identification problem: retain its element E_a , set every other conclusive element to zero, and assign the remaining operator to $E_\perp$. Under the uniform prior, this POVM succeeds with probability arbitrarily close to $p_j^{(a)}/|\Sigma_j|$. Hence

$$p_j^{(a)} \leq |\Sigma_j| s_j.$$

Applying this inequality to each factor in Theorem 1 gives the result. ◀

4 Exponential Separations for Multiparty Index Coordination

We now apply our structural theorems to establish an exponential separation between classical shared randomness and unentangled quantum communication in the multiparty SMP model. To achieve this, we study a relational problem that we call MULTIPARTY INDEX COORDINATION, denoted $IC_{k,n}$.

This problem serves as the natural k -party generalization of the bipartite INDEX COORDINATION problem, which was originally utilized by Gavinsky et al. [6, 7] to demonstrate the incomparability of public coins and quantum messages for two players. In the original two-party setting, Alice and Bob must coordinate to output their respective bits for a specific index known only to Bob. In our multiparty extension, the first $k - 1$ players each receive an n -bit string. The k -th player receives an n -bit string alongside a selector string s with a Hamming weight of exactly $n/2$, which acts as a mask indicating the valid target indices. The goal of the referee is to successfully output the players' bits for any one of these valid indices.

The intuition behind this problem captures the essence of distributed coordination bottlenecks. In the classical public-coin model, the players can achieve this task by using the shared randomness to collectively sample a random index. Because the target set has a relative frequency of $1/2$, a constant number of independent samples contains a valid index with high probability, allowing the players to succeed using only logarithmic communication.

In contrast, in the unentangled quantum model without shared randomness, the players lack any mechanism to coordinate. The first $k - 1$ players have no knowledge of the selector string s , meaning they do not know which parts of their own input send for the referee to success. Consequently, their independent quantum messages must inherently carry enough predictive power to answer queries about a large fraction of their inputs simultaneously.

We formalize this problem as follows:

► **Definition 1** (Problem $IC_{k,n}$). *The problem involves $k \geq 2$ players $P_1, \dots, P_k$ and a central referee, where $n \geq 2$ is even.*

- **Input:** *Each player P_j for $j \in \{1, \dots, k - 1\}$ receives a string $x_j \in \{0, 1\}^n$. Player P_k receives a string $x_k \in \{0, 1\}^n$ and a selector string $s \in \{0, 1\}^n$ with Hamming weight exactly $|s| = n/2$.*
- **Output:** *The referee must output a tuple $(i, x_1^{(i)}, \dots, x_k^{(i)})$ for some index i such that $s^{(i)} = 1$.*

4.1 Classical Protocols for $IC_{k,n}$

Public-Coin Upper Bound.

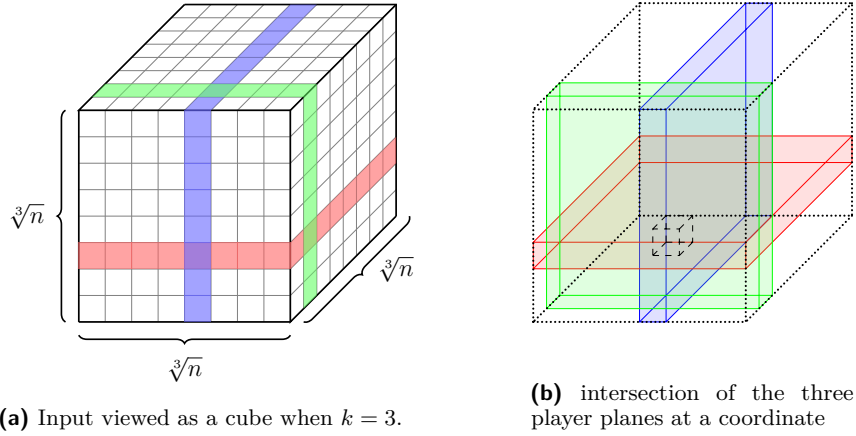
This problem admits a highly efficient classical protocol utilizing shared randomness. The players use the public coin to sample an index $i \in [n]$ uniformly at random. Since the target set has relative frequency $1/2$, a random index is valid with probability $1/2$. All players send their bit $x_j^{(i)}$ and the index i ; player P_k additionally sends $s^{(i)}$. If $s^{(i)} = 0$, the referee aborts ($\perp$). Otherwise, the referee outputs the tuple.

Crucially, this protocol is strictly *unambiguous*: it never outputs an incorrect tuple. To bound the abort probability below a constant $\varepsilon \in (0, 1)$, the players simply repeat this process $O(\log(1/\varepsilon))$ times. Its maximum message length is therefore bounded by

$$R_{\text{pub}}^{\parallel}(IC_{k,n}) = O(\log n).$$

Private-Coin Baseline.

Without public randomness, a simple strategy generalizes the protocol described by Ambainis [1]. Assume first that $n^{1/k}$ is an integer. Each player treats their n -bit input as a flattened k -dimensional tensor of size $n^{1/k} \times \dots \times n^{1/k}$. Each independently selects a uniform random coordinate $c_j \in [n^{1/k}]$ for the j -th dimension and sends c_j together with the corresponding sub-tensor of dimension $k - 1$. Additionally, player P_k sends the corresponding hyperplane of the selector string s . Geometrically, these sub-tensors are orthogonal hyperplanes that intersect in exactly one common index i (see Figure 1). The referee checks whether $s^{(i)} = 1$, outputting the corresponding tuple if so and aborting otherwise. The intersection is uniformly distributed over the n coordinates, so each trial succeeds with probability $1/2$.



■ Figure 1

For arbitrary n , we use a small amount of zero padding and allow slightly different side lengths. Specifically, pad the inputs and selector to length $N = 2^L$, where $L = \lceil \log_2 n \rceil$, and distribute the L binary index coordinates as evenly as possible among the k dimensions. Each side then has length $2^{\lfloor L/k \rfloor}$ or $2^{\lceil L/k \rceil}$, with total volume $N < 2n$; sides of length one are allowed. The arrangement is fixed and known to all parties. Since padded selector positions are zero, they only cause the referee to abort. Each trial still succeeds with probability $n/(2N) \geq 1/4$, and every hyperplane contains at most $2N^{1-1/k} \leq 4n^{1-1/k}$ entries.

Thus, a constant number of independent parallel repetitions suffices for any fixed abort allowance $\varepsilon \in (0, 1)$. The maximum message length is proportional to the size of one hyperplane, with an additional constant factor for the selector. The $O(\log n)$ coordinate headers are absorbed by this bound for $k \geq 2$. Consequently,

$$R_{\text{priv}}^{\parallel}(\text{IC}_{k,n}) = O\left(n^{1-1/k}\right).$$

The protocol is unambiguous, since every conclusive output is valid.

4.2 Bounded-Error Quantum Lower Bound

We first establish our separation in the general bounded-error regime. Deriving a full k -party direct product theorem for bounded-error mixed states is blocked by the complexities of Semidefinite Programming (SDP) duality constraints across multiple tensor products [9].

To circumvent this, we adapt our techniques to utilize the original two-register asymmetric direct product theorem of Gavinsky et al. We conceptually group the first $k - 1$ independent

players into a single “meta-player” by studying the parity of their inputs. This workaround allows us to embed the k -dimensional multiparty state space into a simple 2-register bipartite problem, enabling the use of these established bounded-error results.

► **Lemma 1** (Parity State Factorization [5]). *Let $n \in \mathbb{N}$ and let $X_1, \dots, X_n$ be independent, uniformly distributed random bits. For each $j \in [n]$ and $x \in \{0, 1\}$, let $\rho_j^x \in \mathcal{D}(\mathcal{H}_j)$ be the state of the j -th quantum system conditioned on $X_j = x$. Define the parity variable $U_n = \bigoplus_{j=1}^n X_j$, and let α_n^u be the global state conditioned on $U_n = u$:*

$$\alpha_n^u = \frac{1}{2^{n-1}} \sum_{\substack{x \in \{0,1\}^n \\ \bigoplus_{j=1}^n x_j = u}} \bigotimes_{j=1}^n \rho_j^{x_j}, \quad u \in \{0, 1\}.$$

Then the trace distance between the two parity states factors perfectly:

$$\|\alpha_n^0 - \alpha_n^1\|_{\text{tr}} = \prod_{j=1}^n \|\rho_j^0 - \rho_j^1\|_{\text{tr}}.$$

Proof sketch. For each $j \in [n]$, define $\Delta_j = \rho_j^0 - \rho_j^1$. Subtracting the two conditional mixtures and expanding the tensor product of the local differences gives the exact operator identity

$$\alpha_n^0 - \alpha_n^1 = \frac{1}{2^{n-1}} \bigotimes_{j=1}^n \Delta_j.$$

The claim then follows from the multiplicativity of the Schatten 1-norm under tensor products and the normalized convention $\|A\|_{\text{tr}} = \frac{1}{2} \|A\|_1$. ◀

This exact trace distance factorization allows us to replace the massive global multiparty state property with the product of local capacities, linking our setup to Gavinsky et al.’s 2-register theorem:

► **Proposition 2** (Bounded-Error 2-Register Direct Product [7]). *Let $\alpha_X \otimes \beta_Y$ be a tensor product encoding independent, uniformly distributed random bits X, Y . Let $b = D_\eta(\beta_0, \beta_1)$ be the maximum probability of a (b, η) -predictor for Y . The optimal joint probability p of obtaining a $(p, \eta/2)$ -predictor for the pair (X, Y) is bounded by:*

$$p \leq 32 \|\alpha_0 - \alpha_1\|_{\text{tr}} \cdot b.$$

Now we are ready to prove the communication lower bound for the bounded-error regime.

► **Theorem 2** (Bounded-Error Lower Bound). *For any constant error $\varepsilon < 1/8$, every bounded-error quantum SMP protocol without shared randomness or pre-shared entanglement that solves $\text{IC}_{k,n}$ has maximum message length*

$$m = \max_{j \in [k]} m_j = \Omega\left(n^{\frac{k-1}{k+1}}\right)$$

qubits. Equivalently, at least one player must send this many qubits. The implicit constant depends only on ε and is independent of both n and k .

Proof sketch. Fix a protocol in which every message has at most m qubits. For each of the first $k - 1$ players, the random access code bound limits the total squared distinguishability of the individual input bits. Pooling these quantities lets us choose a selector set of size $n/2$ on which all those players are collectively weak. For an index in this set, we replace their full tuple of bits by its parity: any referee that recovers the tuple also recovers this parity together with the last player's bit. Independence makes the distinguishability of the two parity states equal to the product of the local distinguishabilities, which is can be bounded by AM-GM. The two-register identification bound (Proposition 2) then bounds the probability of jointly predicting the parity and the last bit by this small product times the last player's prediction capacity. A second random access code bound over the information of the k -th player bounds the sum of those capacities over all indices. Summing the resulting success bounds and forcing the constant overall success yields $m = \Omega(n^{(k-1)/(k+1)})$. ◀

Proof. Fix a bounded-error protocol P in which every player sends at most m qubits. We assume $k \geq 2$ and that n is even.

Bounding the Information

For every player $j \in [k-1]$ and coordinate $i \in [n]$, let $\rho_{j,i}^0$ and $\rho_{j,i}^1$ be the average message states conditioned on $x_{j,i} = 0$ and $x_{j,i} = 1$, respectively, and write $d_{j,i} := \|\rho_{j,i}^0 - \rho_{j,i}^1\|_{\text{tr}}$. The Holevo-Helstrom measurement is a $(1, 1/2 - d_{j,i}/2)$ -predictor for $x_{j,i}$. Since $1 - H(1/2 - t/2) \geq c_0 t^2$ for some universal constant $c_0 > 0$ and every $t \in [0, 1]$, the RAC inequality implies

$$\sum_{i=1}^n d_{j,i}^2 \leq C_0 m \quad \text{for every } j \in [k-1],$$

where $C_0 \geq 1$ is a universal constant.

We pool these quantities by defining $c_i := \sum_{j=1}^{k-1} d_{j,i}^2$. Summing the preceding RAC bounds over the first $k - 1$ players gives

$$\sum_{i=1}^n c_i = \sum_{j=1}^{k-1} \sum_{i=1}^n d_{j,i}^2 \leq C_0(k-1)m.$$

Constructing the Hard Distribution

We now use the pooled quantities to fix the selector of player P_k . By Markov's inequality, fewer than $n/2$ coordinates satisfy

$$c_i > 2 \frac{C_0(k-1)m}{n} = \frac{2C_0(k-1)m}{n}.$$

Consequently, there is a subset $S \subseteq [n]$ of size exactly $n/2$ such that $c_i \leq 2C_0(k-1)m/n$ for every $i \in S$. We fix the selector string s to be the indicator of this set.

With s fixed, let

$$\beta_{i,s}^z := \mathbb{E}_{x_k: x_{k,i}=z} [\rho_k(x_k, s)], \quad b_i := D_{4\varepsilon}(\beta_{i,s}^0, \beta_{i,s}^1).$$

Thus b_i is the optimal conclusive probability of a 4ε -error predictor for the i -th bit of x_k . Because $4\varepsilon < 1/2$, the RAC inequality applied to the fixed encoding $x_k \mapsto \rho_k(x_k, s)$ yields

$$\sum_{i=1}^n b_i \leq C_1 m,$$

where $C_1 > 0$ depends only on the fixed error parameter ε , and not on n or k .

Collapsing the First $k - 1$ Players

Fix $i \in S$ and define the parity $U_i := x_{1,i} \oplus \dots \oplus x_{k-1,i}$. Let α_i^0 and α_i^1 be the joint message states of the first $k - 1$ players conditioned on $U_i = 0$ and $U_i = 1$, respectively. By the parity-state factorization lemma and AM–GM,

$$\begin{aligned} \|\alpha_i^0 - \alpha_i^1\|_{\text{tr}} &= \prod_{j=1}^{k-1} d_{j,i} \\ &\leq \left(\frac{1}{k-1} \sum_{j=1}^{k-1} d_{j,i}^2 \right)^{\frac{k-1}{2}} \\ &= \left(\frac{c_i}{k-1} \right)^{\frac{k-1}{2}} \leq \left(2C_0 \frac{m}{n} \right)^{\frac{k-1}{2}}. \end{aligned}$$

Note here that the factor $2C_0$ is independent of k .

Bounding the Referee's Success

Let p_i be the probability that the referee produces a conclusive output with index i . Since the fixed selector is supported on S , every valid output must use an index in S . Call $i \in S$ *good* if, conditioned on producing an output with index i , the reported tuple is correct with probability at least $1 - 2\varepsilon$, and let $G \subseteq S$ be the set of good indices.

Because the protocol is correct with probability at least $1 - \varepsilon$, an averaging argument gives

$$\begin{aligned} 1 - \varepsilon &\leq \sum_{i \in G} p_i + (1 - 2\varepsilon) \sum_{i \in S \setminus G} p_i \\ &\leq 1 - 2\varepsilon + 2\varepsilon \sum_{i \in G} p_i. \end{aligned}$$

Thus $\sum_{i \in G} p_i \geq 1/2$.

For every $i \in G$, run the referee's measurement and return the parity of the first $k - 1$ reported bits together with the reported k -th bit whenever the output index is i ; otherwise return $\perp$. This gives a $(p_i, 2\varepsilon)$ -predictor for $(U_i, x_{k,i})$. Applying the asymmetric two-register direct-product bound from Proposition 2, followed by the parity estimate above, gives

$$p_i \leq 32 \|\alpha_i^0 - \alpha_i^1\|_{\text{tr}} b_i \leq 32 \left(2C_0 \frac{m}{n} \right)^{\frac{k-1}{2}} b_i.$$

Summing over the good indices and using the RAC bound $\sum_i b_i \leq C_1 m$, we conclude that

$$\frac{1}{2} \leq \sum_{i \in G} p_i \leq 32C_1 m \left(2C_0 \frac{m}{n} \right)^{\frac{k-1}{2}}.$$

Wrapping Up the Lower Bound

Rearranging the preceding inequality, there is a constant $C_2 \in (0, 1]$, depending only on ε , such that

$$m^{\frac{k+1}{2}} \geq C_2 \left(\frac{1}{2C_0} \right)^{\frac{k-1}{2}} n^{\frac{k-1}{2}}.$$

Taking the $2/(k+1)$ -th power gives

$$m \geq C_2^{\frac{2}{k+1}} \left(\frac{1}{2C_0} \right)^{\frac{k-1}{k+1}} n^{\frac{k-1}{k+1}}.$$

The first two factors are bounded below uniformly in k . Indeed, $C_2^{2/(k+1)} \geq C_2^{2/3}$, while

$$\left(\frac{1}{2C_0} \right)^{\frac{k-1}{k+1}} \geq \frac{1}{2C_0}.$$

Hence the implicit constant below depends only on the fixed error parameter and remains independent of n and k , even when $k = k(n)$ grows. Therefore,

$$m = \Omega \left(n^{\frac{k-1}{k+1}} \right).$$

◀

4.3 Unambiguous Strong Quantum Lower Bound

We now consider the unambiguous regime as a special case of independent interest. Given the highly restricted nature of this setting, we are able to directly extend the state discrimination theorem for k registers, bypassing the need for the two-register parity-collapse reduction entirely. This allows us to establish that unentangled quantum messages offer no asymptotic advantage over private coins (uniformly in the number of players), achieving a stronger lower bound.

► **Theorem 3** (Unambiguous Lower Bound). *For any fixed constant $\varepsilon \in [0, 1)$, every unambiguous quantum SMP protocol for $\text{IC}_{k,n}$ without shared randomness or prior entanglement and with abort probability at most ε has maximum message length*

$$m = \max_{j \in [k]} m_j = \Omega(n^{1-1/k})$$

qubits. Equivalently, at least one player must send this many qubits. The implicit constant depends only on ε and is independent of both n and k .

Proof sketch. Fix an unambiguous protocol with constant error probability and maximum message length m . For each player and coordinate, let the local capacity be the largest probability of identifying that input bit without error. The zero-error case of the random access code bound says that the sum of a player's local capacities is at most the message length. We pool the capacities of the first $k-1$ players and fix the selector set among coordinates where their total capacity is small as the hard distribution. At any selected coordinate, the unambiguous direct-product theorem bounds the probability of identifying the entire tuple by the product of the local capacities. AM-GM allows to bound the contribution of the first $k-1$ players, while a second random access code bound controls the sum of the last player's capacities. Therefore the protocol's total success rate is at most $2^{2k-1}m^k/n^{k-1}$. Hence constant success probability requires $m = \Omega(n^{1-1/k})$. ◀

The full proof is deferred to Appendix B.

5 Conclusion and Open Problems

In this work, we investigated the multiparty coordination capabilities of the SMP model, establishing that classical shared randomness is exponentially more powerful than unentangled quantum communication. By introducing and analyzing the natural k -party relational problem $\text{IC}_{k,n}$, we proved robust polynomial lower bounds on the longest quantum message. We achieved this via a dual-regime approach: tight matching bounds in the unambiguous setting, supported by an exact subspace factorization theorem for product states, and a polynomial lower bound in the bounded-error setting via a structural parity-collapse technique. For fixed error parameters, the constants in these bounds and in the public-coin upper bound are independent of k . Thus, the exponential separation in maximum message length holds for every integer-valued function $k = k(n) \geq 2$, including arbitrarily growing functions of n .

The matching private-coin upper bound gives this separation a direct resource interpretation. For every fixed positive abort probability, a simple classical protocol using independent private randomness solves $\text{IC}_{k,n}$ unambiguously with maximum message length $O(n^{1-1/k})$, while every unambiguous quantum protocol without shared randomness or prior entanglement requires $\Omega(n^{1-1/k})$ qubits. Consequently, both models have unambiguous communication complexity $\Theta(n^{1-1/k})$. Even arbitrary local quantum encodings and a joint measurement at the referee cannot asymptotically reduce the communication required by the classical hyperplane protocol. Public randomness, however, allows the players to coordinate their choice of a candidate index and solve the same problem unambiguously with only $O(\log n)$ bits per player. Thus, for this relation, quantum communication provides no asymptotic advantage over private randomness in the unambiguous regime, while access to common random choices yields an exponential improvement. Moreover, both quantum lower-bound exponents tend to one as k grows. When $k \geq c \log n$ for any fixed $c > 0$, both lower bounds are $\Omega(n)$ and match the $O(n)$ protocol that sends the complete inputs and selector. Consequently, quantum and classical private-coin communication both have complexity $\Theta(n)$ in the unambiguous and bounded-error regimes in this range, while public randomness still permits maximum message length $O(\log n)$. This tight linear complexity in both regimes shows that the advantage of public randomness persists even as the coordination task becomes maximally costly in the absence of shared resources.

Despite these results, several fundamental questions remain regarding the exact limits of unentangled quantum communication and multiparty state discrimination.

A Multiparty Direct Product Theorem for Bounded-Error Identification

To derive our bounded-error lower bound, we circumvented the lack of a multiparty direct product theorem by employing a parity-collapse reduction. This allowed us to group the first $k - 1$ players into a single meta-register and apply existing two-register asymmetric direct product theorems. Extending such bounds to the identification of all k labels directly could provide another approach to understanding the bounded-error complexity of multiparty coordination.

► **Open problem 1.** *Is it possible to obtain a generalized direct product theorem that directly bounds the identification probabilities of a target sequence encoded across a k -fold tensor product of mixed states in the bounded-error regime?*

Unifying the Lower Bounds Across Error Regimes

For fixed k , an intriguing mathematical gap remains between our two quantum lower bounds: we established a tight $\Omega(n^{1-1/k})$ lower bound for the unambiguous regime, but only an $\Omega\left(n^{\frac{k-1}{k+1}}\right)$ bound for the bounded-error regime. These exponents arise from two different specializations of the RAC inequality. In the unambiguous setting, the zero-error requirement sets $\eta_i = 0$ and yields an L_1 linear constraint on local identification probabilities. In the bounded-error setting, applying RAC to the biases of Helstrom predictors yields an L_2 sum-of-squares constraint on local trace distances. Whether this difference reflects the complexity of the problem or a limitation of the proof remains open.

► **Open problem 2.** *For fixed k , can the bounded-error quantum lower bound for $\text{IC}_{k,n}$ be improved to match the $\Omega(n^{1-1/k})$ unambiguous bound, or does there exist an unentangled quantum protocol that exploits the permitted error margin to achieve $O\left(n^{\frac{k-1}{k+1}}\right)$ communication?*

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A

 Proof of Theorem 1

We first establish an algebraic technical lemma demonstrating that the orthogonal complement of the sum of “wrong” tensor product spaces perfectly factorizes.

► **Lemma 2** (Generalized Subspace Factorization). *Let $k \in \mathbb{N}$. For each $j \in [k]$, let $\mathcal{H}_j$ be a finite-dimensional Hilbert space, and let Σ_j be a finite alphabet. Suppose $\mathcal{H}_j = \sum_{a \in \Sigma_j} S_j^{(a)}$ where $S_j^{(a)} \subseteq \mathcal{H}_j$ are closed subspaces. Here and throughout the statement, a sum of subspaces denotes their linear span and need not be direct. For any target string $b \in \Sigma_1 \times \cdots \times \Sigma_k$, define the global “wrong” subspace W_b as the sum of the product subspaces indexed by label strings $b' \neq b$:*

$$W_b = \sum_{b' \neq b} \bigotimes_{j=1}^k S_j^{(b'_j)}.$$

Then the orthogonal complement of W_b in the full tensor space $\mathcal{H} = \bigotimes_{j=1}^k \mathcal{H}_j$ factorizes completely:

$$W_b^\perp = \bigotimes_{j=1}^k \left(\sum_{a \neq b_j} S_j^{(a)} \right)^\perp.$$

Proof. For each register $j \in [k]$, define the local “wrong” subspace

$$T_{j,b_j} := \sum_{a \neq b_j} S_j^{(a)}.$$

For every $t \in [k]$, we define the local cylinders

$$C_t := \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_{t-1} \otimes T_{t,b_t} \otimes \mathcal{H}_{t+1} \otimes \cdots \otimes \mathcal{H}_k.$$

We first show that

$$W_b = \sum_{t=1}^k C_t.$$

Indeed, every string $b' \neq b$ differs from b in at least one coordinate t . For such a coordinate, $S_t^{(b'_t)} \subseteq T_{t,b_t}$, and hence

$$\bigotimes_{j=1}^k S_j^{(b'_j)} \subseteq C_t.$$

Summing over all $b' \neq b$ gives $W_b \subseteq \sum_{t=1}^k C_t$. Conversely, the assumptions $\mathcal{H}_j = \sum_{a \in \Sigma_j} S_j^{(a)}$ and $T_{t,b_t} = \sum_{a \neq b_t} S_t^{(a)}$, together with distributivity of tensor products over finite sums, imply

$$C_t = \sum_{\substack{a \in \Sigma_1 \times \cdots \times \Sigma_k \\ a_t \neq b_t}} \bigotimes_{j=1}^k S_j^{(a_j)} \subseteq W_b.$$

Therefore, the two subspaces are equal.

We next compute the orthogonal complement of each cylinder C_t . The orthogonal decomposition $\mathcal{H}_t = T_{t,b_t} \oplus T_{t,b_t}^\perp$ induces the orthogonal decomposition

$$\mathcal{H} = C_t \oplus (\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_{t-1} \otimes T_{t,b_t}^\perp \otimes \mathcal{H}_{t+1} \otimes \cdots \otimes \mathcal{H}_k).$$

Consequently,

$$C_t^\perp = \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_{t-1} \otimes T_{t,b_t}^\perp \otimes \mathcal{H}_{t+1} \otimes \cdots \otimes \mathcal{H}_k.$$

Finally, the orthogonal complement of a finite sum of subspaces is the intersection of their orthogonal complements, and therefore

$$W_b^\perp = \left(\sum_{t=1}^k C_t \right)^\perp = \bigcap_{t=1}^k C_t^\perp.$$

For each t , the orthogonal projector onto $C_t^\perp$ is

$$Q_t = I_{\mathcal{H}_1} \otimes \cdots \otimes I_{\mathcal{H}_{t-1}} \otimes \Pi_{T_{t,b_t}^\perp} \otimes I_{\mathcal{H}_{t+1}} \otimes \cdots \otimes I_{\mathcal{H}_k}.$$

These projectors act nontrivially on distinct tensor factors and hence commute pairwise. The product of a finite family of commuting orthogonal projectors is the orthogonal projector onto the intersection of their ranges; see [11, §42] for a proof of this fact. Therefore,

$$\Pi_{W_b^\perp} = \prod_{t=1}^k Q_t = \bigotimes_{j=1}^k \Pi_{T_{j,b_j}^\perp}.$$

Taking ranges on both sides gives

$$W_b^\perp = \bigotimes_{j=1}^k T_{j,b_j}^\perp = \bigotimes_{j=1}^k \left(\sum_{a \neq b_j} S_j^{(a)} \right)^\perp,$$

which concludes the proof. ◀

Now we proceed with the proof of the theorem. Let $S_j^{(a)} = \text{supp}(\alpha_j^{(a)})$. We can restrict each local space to the sum of its supports [18], so $\mathcal{H}_j = \sum_{a \in \Sigma_j} S_j^{(a)}$.

First, consider the local identification problem. To identify the state $\alpha_j^{(b_j)}$ locally without error, a measurement operator E_{b_j} must satisfy $\text{Tr}(E_{b_j} \alpha_j^{(a)}) = 0$ for all incorrect symbols $a \neq b_j$. Since both operators are positive semidefinite, this is equivalent to requiring their supports to be orthogonal. Thus, defining the local “wrong” subspace as $T_{j,b_j} = \sum_{a \neq b_j} S_j^{(a)}$, every feasible operator E_{b_j} satisfies $\text{supp}(E_{b_j}) \subseteq T_{j,b_j}^\perp$. Moreover, because E_{b_j} is a POVM element, $0 \preceq E_{b_j} \preceq I$, and hence $E_{b_j} \preceq \Pi_{T_{j,b_j}^\perp}$. The projector $\Pi_{T_{j,b_j}^\perp}$ is itself feasible: set every other conclusive POVM element to zero and take $E_\perp = I - \Pi_{T_{j,b_j}^\perp}$. Therefore, the optimal local success probability is

$$p_j^{(b_j)} = \text{Tr} \left(\alpha_j^{(b_j)} \Pi_{T_{j,b_j}^\perp} \right).$$

Now, consider the global identification problem. To unambiguously identify the target sequence b , the global POVM element $E_{\text{global}}^{(b)}$ must never trigger if any incorrect product state $\rho_{b'}$ (where $b' \neq b$) was sent. This enforces the strict condition:

$$\forall b' \neq b, \quad \text{Tr} \left(E_{\text{global}}^{(b)} \rho_{b'} \right) = 0.$$

Because $\rho_{b'} = \bigotimes_{j=1}^k \alpha_j^{(b'_j)}$, the support of $\rho_{b'}$ is exactly the tensor product of the local supports: $\bigotimes_{j=1}^k S_j^{(b'_j)}$. For the trace to be zero, the measurement operator $E_{\text{global}}^{(b)}$ must be orthogonal to the support of every single incorrect product state. By extension, it must be orthogonal to the entire subspace spanned by the union of all these incorrect supports.

Let W_b denote this global “wrong” subspace, defined explicitly as the sum of the supports of all incorrect product states:

$$W_b = \sum_{b' \neq b} \text{supp}(\rho_{b'}) = \sum_{b' \neq b} \bigotimes_{j=1}^k S_j^{(b'_j)}.$$

Consequently, the support of $E_{\text{global}}^{(b)}$ must be contained in $W_b^\perp$. Since $0 \preceq E_{\text{global}}^{(b)} \preceq I$, it follows that $E_{\text{global}}^{(b)} \preceq \Pi_{W_b^\perp}$. Therefore, the maximum global success probability is bounded by

$$P(b) \leq \text{Tr}(\rho_b \Pi_{W_b^\perp}).$$

At this point, we apply our Generalized Subspace Factorization (Lemma 2), which establishes that the orthogonal complement of this global sum of product subspaces factorizes into the tensor product of the local orthogonal complements:

$$W_b^\perp = \bigotimes_{j=1}^k T_{j,b_j}^\perp.$$

Substituting this factorization into the preceding trace bound and using multiplicativity of the trace over tensor products, we obtain:

$$P(b) \leq \text{Tr} \left(\left(\bigotimes_{j=1}^k \alpha_j^{(b_j)} \right) \bigotimes_{j=1}^k \Pi_{T_{j,b_j}^\perp} \right) = \prod_{j=1}^k \text{Tr} \left(\alpha_j^{(b_j)} \Pi_{T_{j,b_j}^\perp} \right) = \prod_{j=1}^k p_j^{(b_j)}.$$

This completes the proof.

B Proof of Theorem 3

Proof. We assume there exists an unambiguous protocol for the problem $\text{IC}_{k,n}$. Let $m_j \leq m$ be the message size in qubits sent by player P_j . For each player P_j with $j \in [k-1]$ and coordinate $i \in [n]$, let $\rho_{j,i}^0$ and $\rho_{j,i}^1$ be the average message states conditioned on $x_{j,i} = 0$ and $x_{j,i} = 1$, respectively. We denote by $a_j^{(i)}$ the optimal average success probability, under the uniform prior on the bit, of a single unambiguous POVM for this binary ensemble, as in Corollary 1.

Bounding the Information

For every coordinate $i \in [n]$, the optimal unambiguous measurement for player P_j , where $j \in [k-1]$, is an $(a_j^{(i)}, 0)$ -predictor: it produces a conclusive answer with probability $a_j^{(i)}$ and, conditioned on doing so, never outputs the wrong bit. Applying the RAC inequality (Proposition 1) with $\lambda_i = a_j^{(i)}$ and $\eta_i = 0$ gives

$$\sum_{i=1}^n a_j^{(i)} (1 - H(0)) \leq m_j.$$

Since $H(0) = 0$, this takes the simple form

$$\sum_{i=1}^n a_j^{(i)} \leq m_j.$$

We aggregate the capacities of the first $k-1$ players. Let $c_i := \sum_{j=1}^{k-1} a_j^{(i)}$ represent their pooled unambiguous prediction capacity for the i -th coordinate, and let $M := \sum_{j=1}^{k-1} m_j \leq (k-1)m$ be their total communication budget. Summing the individual RAC inequalities yields

$$\sum_{i=1}^n c_i = \sum_{j=1}^{k-1} \sum_{i=1}^n a_j^{(i)} \leq \sum_{j=1}^{k-1} m_j = M.$$

Constructing the Hard Distribution

Using this aggregated capacity, we construct a hard input distribution by fixing the k -th player's selector string s . Specifically, we require a subset S of size exactly $n/2$ where the pooled capacity c_i is small. If $M = 0$, then every c_i is zero and any subset $S \subseteq [n]$ of size $n/2$ has the required property. Otherwise, by Markov's inequality, the number of indices $i \in [n]$ exceeding a threshold T is strictly bounded by M/T . To guarantee there are at least $n/2$ "good" indices remaining, we must restrict the maximum number of "bad" indices to be strictly less than $n - n/2 = n/2$. Setting $M/T = n/2$ yields the precise threshold $T = 2M/n$. Therefore, there exists a static subset $S \subseteq [n]$ of size $n/2$ such that for all $i \in S$, $c_i \leq 2M/n$. We fix the hard instance distribution by setting s to be the indicator vector of this specific set S .

With this selector now fixed, let $\rho_{k,i}^0$ and $\rho_{k,i}^1$ be the average message states of player P_k on inputs (x_k, s) , conditioned on $x_{k,i} = 0$ and $x_{k,i} = 1$, respectively. Let $a_k^{(i)}$ be the optimal average success probability of a single unambiguous POVM for this binary ensemble. Applying the RAC inequality to the encoding $x_k \mapsto \rho_k(x_k, s)$ for this fixed selector gives

$$\sum_{i=1}^n a_k^{(i)} \leq m_k.$$

Bounding the Referee Prediction

Let p_i be the unconditional probability, over uniformly random input strings and the protocol's private randomness, that the referee outputs a valid tuple with index i . Because the protocol is strictly unambiguous, it is forbidden from outputting an incorrect tuple; moreover, the selector string s is supported on S , so a valid answer must use an index in S . Consequently, the total probability of generating a valid tuple is $\sum_{i \in S} p_i$.

For a fixed $i \in S$ and a fixed bit tuple $b \in \{0, 1\}^k$, independence of the players' inputs and the product-message assumption imply that the joint average message conditioned on $(x_{1,i}, \dots, x_{k,i}) = b$ is the product state $\bigotimes_{j=1}^k \rho_{j,i}^{b_j}$. The POVM element that outputs (i, b) identifies this tuple unambiguously. Applying Corollary 1 and then averaging over the uniformly distributed tuple b gives

$$p_i \leq 2^k a_k^{(i)} \prod_{j=1}^{k-1} a_j^{(i)}.$$

Therefore, the required global success probability implies

$$1 - \varepsilon \leq \sum_{i \in S} p_i \leq 2^k \sum_{i \in S} \left(a_k^{(i)} \prod_{j=1}^{k-1} a_j^{(i)} \right).$$

To bridge the product of local capacities to our pooled sum c_i , we apply the Arithmetic Mean-Geometric Mean (AM-GM) inequality to the first $k-1$ players. By isolating the product and raising both the arithmetic and geometric means to the power of $k-1$, we obtain:

$$\prod_{j=1}^{k-1} a_j^{(i)} \leq \left(\frac{1}{k-1} \sum_{j=1}^{k-1} a_j^{(i)} \right)^{k-1} = \left(\frac{c_i}{k-1} \right)^{k-1} \leq \left(\frac{2M}{(k-1)n} \right)^{k-1}.$$

Wrapping Up the Lower Bound

Substituting this uniform bound back into the global success sum yields:

$$1 - \varepsilon \leq 2^k \sum_{i \in S} a_k^{(i)} \left(\frac{2M}{(k-1)n} \right)^{k-1} \leq 2^k m_k \left(\frac{2M}{(k-1)n} \right)^{k-1}.$$

Finally, we evaluate this expression asymptotically. Substituting $M \leq (k-1)m$ and $m_k \leq m$ gives:

$$1 - \varepsilon \leq 2^k m \left(\frac{2m}{n} \right)^{k-1}.$$

Rearranging the terms to isolate m^k yields

$$m^k \geq (1 - \varepsilon) 2^{-k} \left(\frac{1}{2} \right)^{k-1} n^{k-1}.$$

Taking the k -th root gives the explicit bound

$$m \geq \frac{1}{2} (1 - \varepsilon)^{1/k} \left(\frac{1}{2} \right)^{\frac{k-1}{k}} n^{1-\frac{1}{k}}.$$

For every $k \geq 2$, we have $(1 - \varepsilon)^{1/k} \geq (1 - \varepsilon)^{1/2}$ and

$$\left(\frac{1}{2} \right)^{\frac{k-1}{k}} \geq \frac{1}{2}.$$

Consequently,

$$m \geq \frac{\sqrt{1-\varepsilon}}{4} n^{1-\frac{1}{k}} = \Omega \left(n^{1-\frac{1}{k}} \right),$$

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